

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2019 – 2020 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. EEE

Course Code & Title: EE8691 Embedded Systems

Name of the Faculty member: S. Sharmila Kumari

ACTIVITY

Topic: Watch Dog Timer, Real Time Clock, In Circuit Emulator

Goal (Learning Outcome):

- ❖ The students will be able to compare Watch Dog Timer, Real Time Clock and In Circuit Emulator – (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Mini - map

Justification for Choosing the Activity for the topic:

1. Differentiate Watch Dog Timer, Real Time Clock and In Circuit Emulator

After teaching the concept, I thought of conducting this activity for making the students to give the difference between the three concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, the students were made to pair with their neighbours

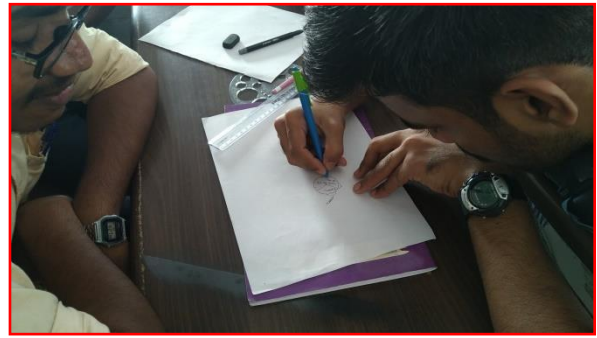
Photographer: one student - Myself

Reporter: Myself

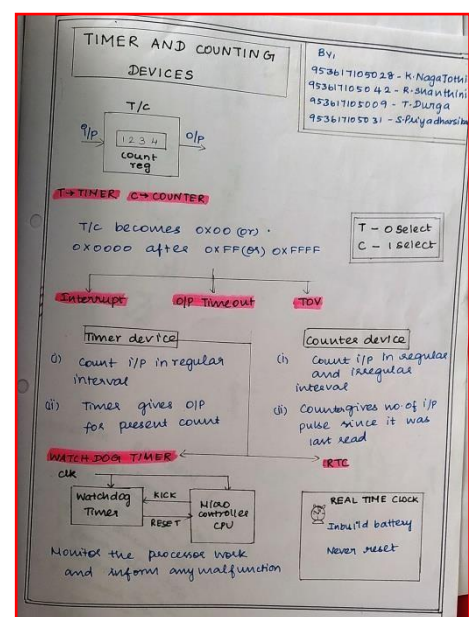
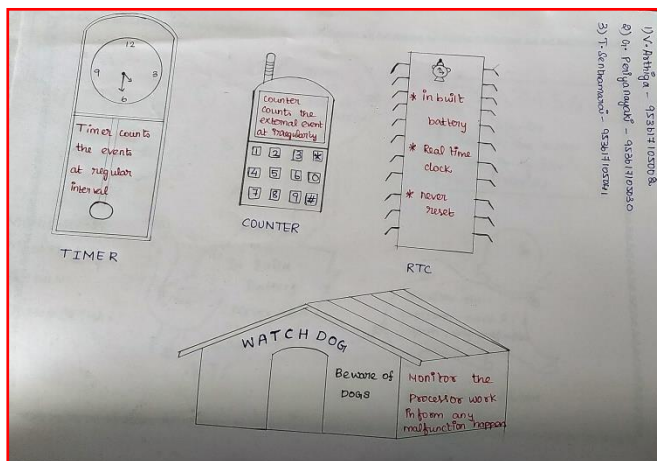
At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about Watch Dog Timer, Real Time Clock and In Circuit Emulator concept for 2 minute.
- ✓ Then I told them to Pair with their neighbours and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design mini-map and submit within 3 minutes.

A view of implementation of the Activity



Samples:



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the concepts of Watch Dog Timer, Real Time Clock and In Circuit Emulator.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 50% of students answered.**

Reflective Critique:

1. Pre-implementation Reflection :

I preferred this Activity because they have Watch Dog Timer, Real Time Clock and In Circuit Emulator separately.

Challenges anticipated:

- ❖ In the class mostly boys hastate to answer to the questions.
- ❖ Time utilization for conducting activity.

Steps taken:

- ❖ While using this activity, the students will get clarity in the timer and counter concept.
- ❖ The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. Post implementation Reflection:

• Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• Challenges:

- Slow learners were not able to understand some topics during discussion hours.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C311.1	2	2	2	1	1	2	-	1	-	1	-	-

References:

1. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps>

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ACTIVITY

Topic: Serial Bus Communication Protocol

Goal (Learning Outcome):

- ❖ The students will be able to compare RS 232, RS 422 and RS 485 – (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Visual Clicker

Justification for Choosing the Activity for the topic:

2. Differentiate RS 232, RS 422 and RS 485

The topic Serial Bus Communication Protocols (RS 232, RS 422 and RS 485) are used for the serial communication purpose. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the three concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept of RS 232, RS 422 and RS 485, the students were made to pair with their neighbours

Total Strength is 55, Number of Pairs – minimum 6 in one team

Photographer: one student - Mr. K. Vignesh (interested in photography)

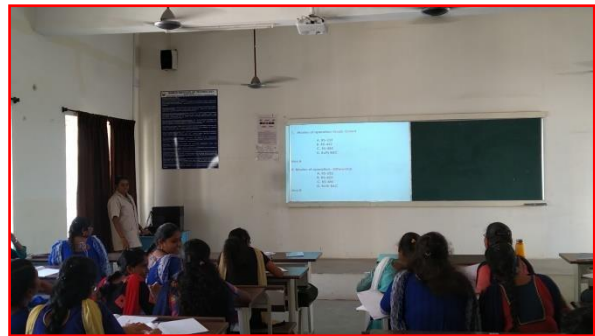
Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about RS 232, RS 422 and RS 485 concept for 1 minute.
- ✓ Then I told them to Pair with their neighbours and discuss for another 1 minute.

- ✓ With the help of PPT prepared (Multiple choice questions) students use their clickers to respond to the question and then the answers were be projected to the students.
- ✓ After completion of the class I uploaded the PPT in the course website (Canvas)

A view of implementation of the Activity



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the concepts of RS232, RS422 and RS485.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 50% of students answered.**

Reflective Critique:

3. Pre-implementation Reflection :

I preferred this Activity because they have RS232, RS422 and RS485 separately.

Challenges anticipated:

- ❖ Time utilization for conducting activity.

4. Post implementation Reflection:

• Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• Challenges:

- Student clicker's were not clear in photos. So clicker tools prepared from students were not advisable.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C311.2	2	3	2	1	1	2	-	1	-	1	-	1

References:

1. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps>

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ACTIVITY

Topic: Mailbox and Pipes

Goal (Learning Outcome):

- ❖ The students will be able to compare mailbox and pipes – (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: One minute paper

Justification for Choosing the Activity for the topic:

Differentiate mailbox and pipe

After teaching the concept, I thought of conducting this activity for making the students to give the difference between mailbox and pipe which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 3 Minutes

Effective presentation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 55

Reporter: Myself

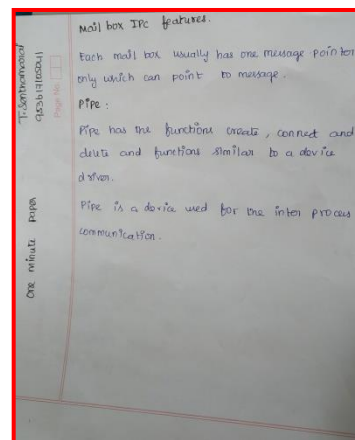
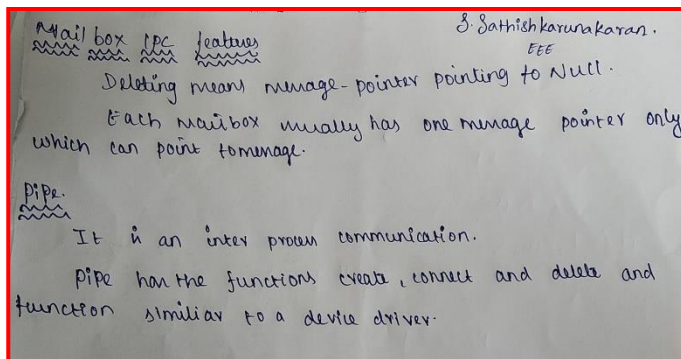
At the end the Class (Last 3 minutes)

- ✓ I asked the students to think about Mailbox and pipes from interprocess communication for 2 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)

A view of implementation of the Activity



Samples:



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the concepts of mailbox and pipes.

Reflective Critique:

• Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C311.4	2	2	2	-	-	-	-	-	-	1	-	1

References:

1. <https://omerad.msu.edu/27-teaching/177-minute-papers>
2. Shibu. K.V, "Introduction to Embedded Systems", 2e, Mc graw Hill, 2017.

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Name of the Faculty member: S.Sharmila Kumari

ACTIVITY

Topic: Object Oriented Model

Goal (Learning Outcome):

- ❖ The students will be able to compute Embedded development models
- ❖ The students will be able to compare Object Oriented Model, State machine model, Concurrent model and Sequential model – (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Think Pair Share

Justification for Choosing the Activity for the topic:

1. Differentiate Object Oriented Model, State machine model, Concurrent model and Sequential model

The topic Object Oriented Model, State machine model, Concurrent model and Sequential model are used for the same purpose. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the different models concepts which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, the students were made to pair with their neighbours (i.e) In a Class.

Total Strength is 55, Number of Pairs – 27

Photographer: one student - Mr.K.Vignesh (interested in photography)

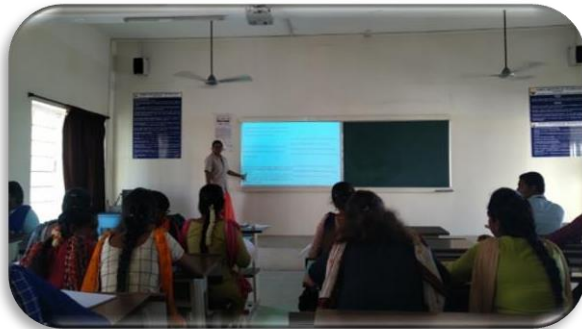
Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to **think** about Object Oriented Model and other model concept for 2 minute.
- ✓ Then I told them to **Pair** with their neighbours and discuss about the difference between the models for another 1 minute.

- ✓ Finally, I selected on Pair from each column randomly and ask them to **share** one difference between the concepts. (3 minutes)
- ✓ When each pair told the points simultaneously, I wrote the points on the board. Finally, I summarized the points and I told one missed point to all the students.
- ✓ With the help of PPT prepared (first slide- Activity name, Second slide-Topic for Think-pair-share, Third slide - answer) the answers were be projected to the students.
- ✓ After completion of the class I uploaded the PPT in the course website (Canvas)

A view of implementation of the Activity while the students are sharing the points



Samples:

THINK PAIR SHARE Embedded kids.

members
MITHINESH G
SHIVAGANESH M

Data flow model:
The Data flow graph is visual model in which the operation on data is represented using a block and data flow is represented using arrows.
Types: i) ADFG, ii) CDFG

State Machine model:
i) Used for Event driven Embedded Systems.
ii) State Machine model describe the System behaviour with following elements.
States: State is a representation of Current Events, Actions, Transition

Sequential Model:
The functions of Sequential model is processing requirements are executed in Sequence

Concurrent Model:
This model is used for Capturing Concurrent executing tasks/process requirements.

Embedded System

Team name: EDLC Queen

P. Sangeetha
(953617105037)
S. Jda Rainy
(953617105016)

Different type of model

- 1) Data flow graph model: Directed graph that show the data dependencies between number of function
- 2) State Machine model: output depends on the entire history of their input and not just on the most recent input. It have a performance that is determined by its history.
- 3) Sequential program model: It is named as linear Programming model. It perform multifunction sequentially and is composed from four software component. Initialization, Reading input, processing & writing output
- 4) Concurrent model: It has multi task that are linked to run these program.
- 5) Object oriented model: Object using a collection of object. It has different classes these are

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that one point was missed with respect to method of computation and finally I insisted the point they missed to tell.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test II – Around 60% of students answered correctly.**

Reflective Critique:

1. Pre-implementation Reflection :

I preferred this Activity because they have Object Oriented Model, State machine model, Concurrent model and Sequential model separately.

Challenges anticipated:

- ❖ In the class mostly boys hastate to answer to the questions.
- ❖ Time utilization for conducting activity.

Steps taken:

- ❖ While using this activity, the students will get clarity in the software development model concept.
- ❖ The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. Post implementation Reflection:

• Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

• Challenges:

- Slow learners were not able to understand some topics during discussion hours.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
C312.3	2	2	3	-	1	-	-	-	-	1	-	1

References:

1. <http://www.readwritethink.org/professional-development/strategy-guides/using-think-pair-share-30626.html>
2. <https://in.video.search.yahoo.com> – Think pair share